

The Last Mile Challenge: Uncovering Delivery Delays, Bottlenecks, and Operational Risks Through Data Analytics

EXECUTIVE INSIGHTS REPORT

Executive Summary

This project analyzed 9,951 shipment records to identify operational bottlenecks, delivery performance issues, regional inefficiencies, seasonal trends, and category-level risks within a logistics network.

The analysis was conducted using SQL for data preparation and business analysis, followed by Python-based visualizations for executive reporting.

The objective was to identify areas where operational improvements could reduce delivery delays, minimize service-level agreement (SLA) violations, and improve overall logistics performance.

Business Context

In logistics operations, delivery delays directly impact customer satisfaction, operational costs, and service-level commitments.

Management requires visibility into:

- Which locations create the highest operational pressure.

- Which periods of the year experience elevated delays.

- Which product categories contribute most to delivery failures.

- Whether shipment characteristics influence performance.

- Which operational areas should be prioritized for improvement.

This analysis was designed to answer these questions using historical shipment data.

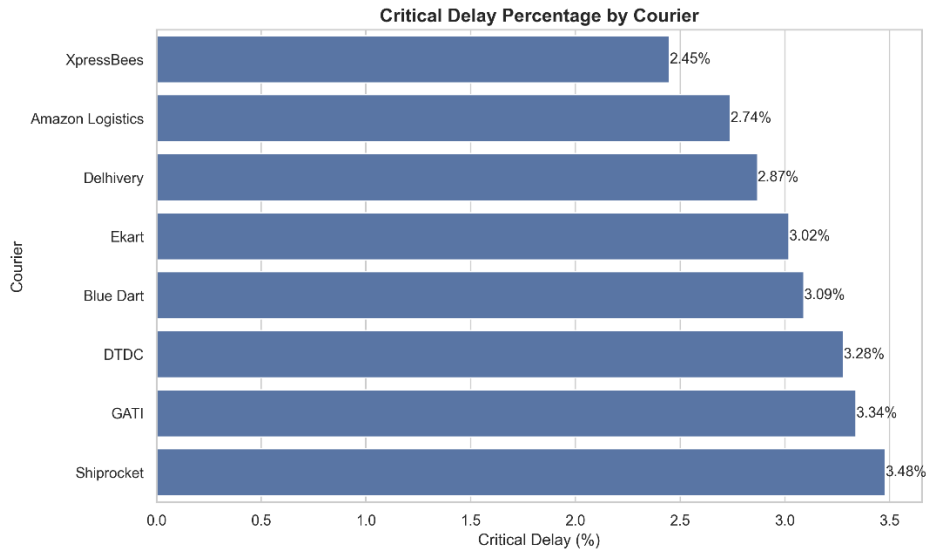
Key Business Findings

A. Regional Bottlenecks

Analysis revealed that Bangalore handled the highest shipment volume in the network, making it the most operationally significant city.

Although Kochi recorded the highest average delay, its lower shipment volume reduced its overall business impact.

Delhi and Hyderabad emerged as higher-priority concerns because they combined elevated delays with significantly larger shipment volumes.



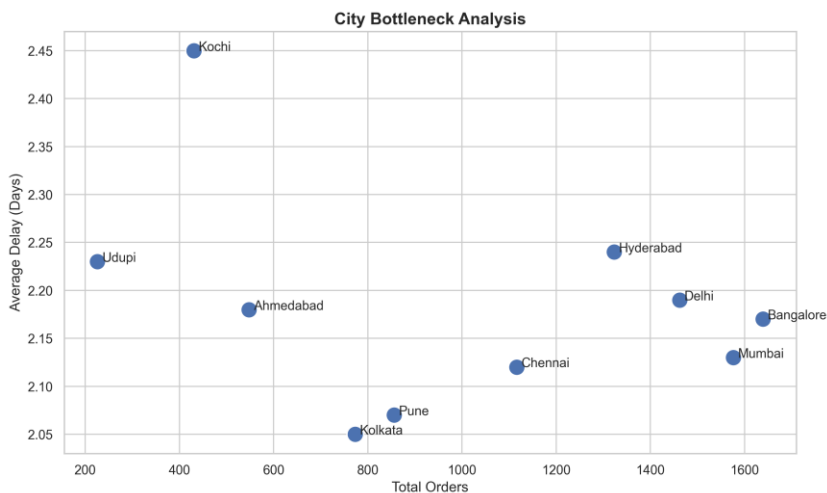
Business Impact

High-volume cities experiencing elevated delays can create widespread customer dissatisfaction and increase operational costs.

B. Zone-Level Performance

North and West zones demonstrated comparatively higher delivery delays.

Central zone processed the largest shipment volume, indicating that operational pressure is concentrated within a few key regions.



Business Impact

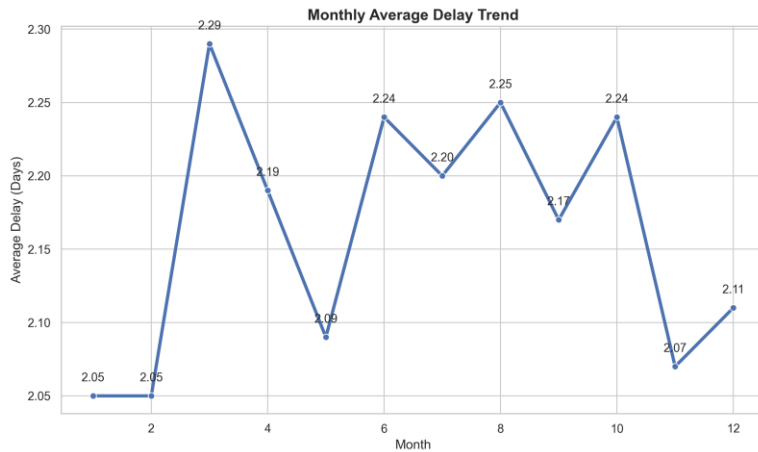
Targeted process improvements within high-delay zones could significantly improve overall network performance.

C. Seasonal Risk Periods

Monthly trend analysis identified March, August, and October as the highest-risk operational months.

August recorded the highest shipment volume, while October recorded the highest number of critical delivery delays.

The period between June and October consistently demonstrated elevated delay levels.



Business Impact

Additional logistics capacity and proactive planning may be required during peak operational periods.

D. Product Category Risks

Electronics emerged as the most business-critical category.

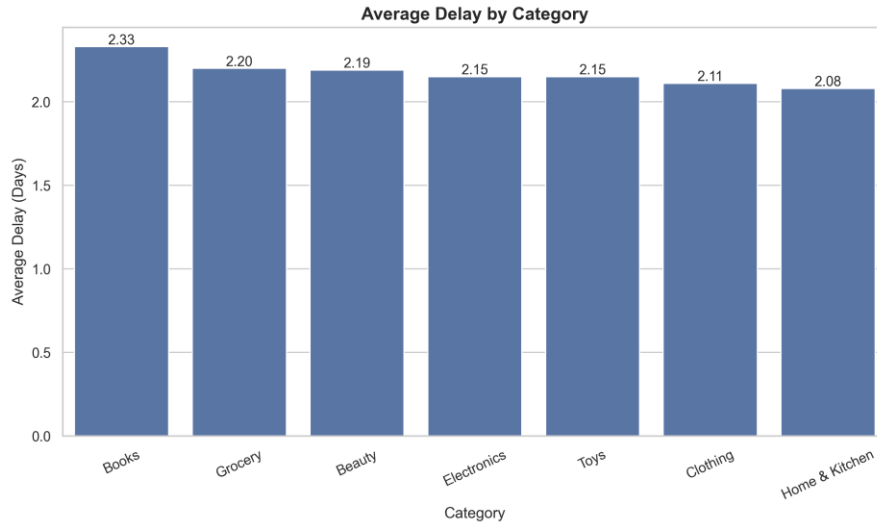
Key observations:

- Highest shipment volume.

- Highest number of critical delays.

- Significant contribution to overall operational workload.

Books recorded the highest average delivery delay, suggesting category-specific delivery challenges.



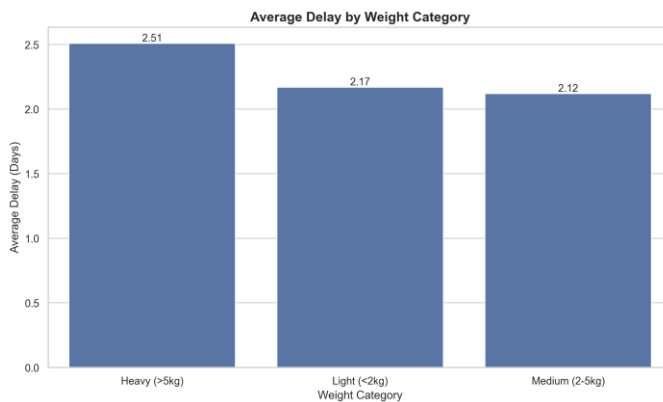
Business Impact

Improving delivery performance within Electronics can generate the greatest overall business benefit.

E. Shipment Weight Analysis

Heavy shipments recorded the highest average delay.

However, light-weight shipments contributed the majority of critical delays due to their dominant share of shipment volume.



Business Impact

Operational focus should prioritize shipment volume rather than weight alone.

Strategic Recommendations

Based on the analysis, the following actions are recommended:

Prioritize operational improvements in Bangalore and Delhi.

Increase logistics capacity during the June–October period, particularly in August and October.

Establish enhanced SLA monitoring for Electronics shipments.

Investigate recurring causes of slight delivery delays, which represent the majority of delivery failures across all courier partners.

Develop regional performance dashboards for high-delay zones to enable proactive intervention.

Expected Business Benefits

Implementation of these recommendations can contribute to:

- Improved delivery performance.
- Reduction in critical delivery failures.
- Better resource allocation during peak periods.
- Enhanced customer satisfaction.
- Improved operational visibility for decision-makers.

Conclusion

The analysis identified a combination of geographical, seasonal, and product-related factors influencing delivery performance.

The most significant opportunities for improvement were concentrated in high-volume cities, peak operational months, and the Electronics category.

These findings provide a data-driven foundation for operational planning, performance monitoring, and future logistics optimization initiatives.